

Haobo Zhao

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Education

Johns Hopkins University , Baltimore, MD	Sep. 2023 - Present
• <i>Master in Mechanical Engineering</i> • Advisors: Dr. Rajat Mittal and Dr. Jung-Hee Seo • Coursework: CFD, Fluid Dynamics I & II, Convection, Numerical Method, Turbulence, Multiphase Flow	GPA: 3.86/4.0
Southern Illinois University , Carbondale, IL	2022 - 2023
• <i>B.S. in Aviation Technologies</i> (Dual Degree with SAU) • Graduated Magna cum Laude, Dean's List: Fall 2022, Fall 2023	GPA: 3.81/4.0
Shenyang Aerospace University , Shenyang, China	2019 - 2023
• <i>Bachelor in Aircraft Propulsion Engineering</i> • National Scholarship (2021, top 1% in Department) • SAU First Class Scholarship (Fall 2020, Fall 2021, Spring 2022)	GPA: 3.8/4.0

Research

Johns Hopkins University , Baltimore, MD	Jan. 2024 - Present
• Master Thesis (Advisors: Dr. Rajat Mittal, Dr. Jung-Hee Seo) – Department of Mechanical Engineering • Developed and validated a patient-specific <i>CFD</i> analysis procedure for the pancreatic duct from <i>MRI</i> data. • Collaborated with the JHU medical team led by Dr. Akshintala to find compare with clinical data. • Formulated and implemented a theoretical flow model to predict pressure variations along the PD. • Conducted geometry analysis for the theoretical flow model and implemented geometry analysis toolkits, including <i>VMTK</i> and <i>NeuroMorph</i>. • Contributed to debugging and enhancing the <i>NeuroMorph</i> toolkit.	

Publications and Presentations

Conference Presentations

- Zhao, H., Seo, J. H., Akshintala, V., Boparai, I., & Mittal, R. (2024). Computational Modeling of Pancreatic Duct Flows for a Novel Non-invasive Diagnosis of Chronic Pancreatitis. *Bulletin of the American Physical Society*.

In Preparation

- Zhao, H., Seo, J. H., Akshintala, V., Boparai, I., & Mittal, R. (2024). **Non-invasive assessment of pancreatic duct hypertension using computational flow modeling**. (To be submitted to the *Journal of Biomechanical Engineering*).

Experience

CFD Visualization Engineer , CertAir LLC	Oct. 2024 - Present
• Conducted CFD simulations of cleanroom airflow to optimize layout and design for controlled environments. • Developed and implemented visualization techniques for airflow patterns and pressure distribution to diagnose cleanroom performance issues. • Automated processes for robotic pressure measurement, particle detection, and report generation, streamlining cleanroom certification workflows. • Collaborated with multidisciplinary teams to enhance cleanroom compliance with ISO standards and improve	

operational efficiency.

Skills

Programming: Python, MATLAB, C++

Software: ANSYS CFX, Fluent, COMSOL, Mimics, 3-Matic, Blender, AutoCAD, SOLIDWORKS

Tools: Git, $\text{\LaTeX}$

Honors and Awards

- **National Scholarship (2021):** Awarded to top 1% in department for academic excellence.
- **First Prize, National Mathematics Competition (China, 2020):** Top 8% of participants.
- **Third Prize, Mechanics Competition of Zhou Peiyuan (China, 2021):** Recognized for excellence in mechanics.
- **Top 5 in China, iCAN Innovation Contest (2021):** Ranked 5th out of 3000 teams nationally (Group Award).

Volunteering & Teaching

Student Teacher, Shenyang Aerospace University Dec. 2019 - Dec. 2021

- Tutored Calculus I & II and College Physics I, including study sessions and final exam reviews.
- Assisted 66 students, improving understanding and academic performance.
- Created final review materials, including mind maps and instructional videos to aid retention.

Core Team Member, SIU-SAU University Collaboration Program May 2022 - Dec. 2022

- Counseled students from Shenyang Aerospace University in the SIU-SAU Collaboration program.
- Negotiated with SIU's Center for International Education for course arrangements.
- Provided emotional, mental, and financial support in collaboration with professional counselors.

Other Rewards

Mathematics:

- First Prize, National Mathematics Competition for College Students (Top 8%), 2020
- First Prize, Mathematical Modeling Competition in Liaoning Province, 2021
- Second Prize, Mathematical Modeling Competition in Liaoning Province, 2020
- Second Prize, National Mathematics Competition for College Students, 2021
- Third Prize, China CUMCM (Contemporary Undergraduate Mathematical Contest in Modeling), 2021
- Third Prize, MathorCup Undergraduate Mathematical Modeling Challenge, 2021
- Second Prize, Asia Pacific Mathematical Contest in Modeling, 2020
- Second Prize, Mathematical Contest in Modeling of Three Provinces in Northeast China, 2020

Physics:

- Third Prize, Mechanics Competition in Honour of Zhou Peiyuan, 2021
- First Prize, Physics Experiment Competition in Liaoning Province, 2020

Innovation and Design:

- iCAN Innovation Contest (Finalist), Top 5 in China, 2021
- Third Prize, China College Students' "Internet+" Innovation and Entrepreneurship Competition, 2021
- Second Prize, The 6th China International "Internet+" College Student Innovation and Entrepreneurship Competition SAU Selection, 2020
- First Prize, SAU Future Engine Design Competition